AP CALCULUS AB

Unit 1 • Lesson 1-9 • Guided notes

Lesson 1-9 • IVT, EVT, and MVT

Guided notes • Fill in the blanks and work the examples in the boxes

Objective: know when to use IVT, EVT and MVT, and justify the answer.

Table review

	IVT	EVT	MVT
Needs			
Proves			

Remember: differentiable $\Rightarrow$ _____, but continuous does _____ imply differentiable.

Example 1

f is differentiable for all x.

x	1	3	5	8
$f(x)$	4	-2	6	10

a) Must there be a c in $(1,3)$ with $f(c) = 0$? b) Must there be a c in $(3,5)$ with $f'(c) = 4$? c) Must f have an absolute maximum on $[1,8]$?

Example 2

$g(x) = \dfrac{1}{x-2}$ on $[1,3]$, with $g(1) = -1$ and $g(3) = 1$. Does the IVT guarantee a c with $g(c) = 0$?

Example 3

h is continuous on $[0,4]$, but its graph has a sharp corner at $x = 2$. Which theorems apply to h on $[0,4]$?

Summary	